

Stablecoin StressBench: Measuring the 12× Gap Between Visible and Executable Arbitrage

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Abstract

A price chart shows twelve times more profitable stablecoin arbitrage than a trader can capture. No prior benchmark provides per-minute executable labels, route-level depth provenance, or a hindsight oracle ceiling—every published performance number therefore measures models against windows that cannot be traded at a profit. We introduce Stablecoin StressBench: an 18-event stress catalogue across seven mechanisms (2020–2023), tiered data governance (Tier A: execution-grade; Tier B: price/liquidity; Tier C: taxonomy), per-minute VWAP book-walking labels with route provenance, and a hindsight oracle ceiling. Across 25+ model configurations spanning price rules, tabular classifiers, sequence models, class-balanced variants, and RL, every calm-trained model fails economically; cross-mechanism meta-labeling trained on Terra/LUNA recovers 51.0% of the oracle on SVB (+82.5 bps), and a four-event pooled protocol reaches 51.8%. A conditioned RL agent at identical positive-label density earns −29.2 bps, isolating explicit binary supervision as the ingredient that reward optimisation cannot replicate at low positive rates. We release the catalogue, labels, oracle, and model harness under CC-BY 4.0.

Keywords

stablecoin arbitrage, market microstructure, execution-aware labels, financial benchmarks, cryptocurrency

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1 Introduction

Stablecoin stress is not one thing. From 2020 to 2023 we identify 18 distinct episodes spanning seven mechanism classes: algorithmic death spirals (Terra/LUNA, IRON/TITAN), fiat-reserve shocks (USDC/SVB), regulatory winddowns (BUSD), exchange-credit collapses (FTX, Celsius), DeFi pool imbalances (Curve/USDT), collateral liquidations (DAI Black Thursday), and RWA/niche failures (USDR, Acala). These events differ in trigger, duration, venue, depth regime, and data observability—a Terra spiral looks nothing like an SVB bank-run on a price chart, and both look nothing like a regulatory winddown. Every prior stablecoin stress study examines at most one of these mechanisms; none provide per-minute executable labels or a hindsight oracle ceiling across the space. The benchmark problem is therefore not merely to detect a depeg, but to establish which historical stress signatures remain economically executable, and which transfer across mechanisms.

The answer is stark: **executable arbitrage is microstructure-scarce even when visible arbitrage is abundant**. When SVB

collapsed in March 2023, USDC briefly depegged and cross-venue price gaps appeared profitable on every chart—but only 2.88% of one-minute windows survived VWAP slippage, taker fees, and settlement latency. We call dislocations that appear on price charts but vanish under execution **optical**, and the surviving fraction **executable**. The gap is 12× at retail size and collapses to near-zero at institutional scale. The only path to capturing executable dislocations is training on a mechanistically distinct stress event: a model trained on Terra/LUNA’s algorithmic collapse transfers to SVB’s reserve-bank shock because CEX market makers withdraw depth under uncertainty regardless of the trigger’s cause.

Stablecoin StressBench provides the empirical foundation for both findings: an 18-event source-tracked catalogue, a mechanism taxonomy, a tiered data governance system, VWAP execution labels, and a hindsight oracle ceiling. The remainder of the paper answers four research questions: what the full stress universe looks like (§3.3), how much apparent arbitrage survives execution (§6.1), whether mechanism diversity helps transfer (§6.3), and why reward-only RL fails (§6.3.2).

Contributions.

- (1) **We provide** the first execution-aware stablecoin benchmark: an 18-event source-tracked catalogue across seven mechanisms, tiered data governance, per-minute VWAP labels with route provenance, and a hindsight oracle ceiling. (§3–5)
- (2) **We show** a 12× optical-to-executable gap and that cross-mechanism meta-labeling generalises reliably: training on Terra/LUNA achieves +82.5 bps (51.0% oracle capture, the fraction of the oracle’s per-trade return recovered) on SVB, and pooling four events across three mechanism classes reaches 51.8%, establishing mechanism-independent transfer saturating near 50%. (§6, Tables 6, 10)
- (3) **We establish** that explicit binary supervision—not label density—is the binding constraint: a conditioned PPO-GRU at 17% positive density earns −29.2 bps versus meta-labeling’s +82.5 bps at identical density. (§6, Table 10)
- (4) **We identify** a structural ~50% ceiling on oracle capture that is invariant to mechanism diversity: late-stage microstructure ambiguity degrades the depth-withdrawal signature before dislocation closes, bounding deployable performance regardless of training breadth. (§6.3)

2 Related Work

Uhlig [1] and Griffin and Shams [2] analyse Terra/LUNA and stablecoin non-fundamental demand. We use Terra/LUNA as a cross-mechanism training split because its algorithmic trigger is mechanistically distinct from SVB’s fiat-reserve failure. Hernandez Cruz et al. [21] study DEX liquidity during the SVB collapse without modelling CEX execution costs. No prior work provides per-minute executable labels or a hindsight oracle ceiling.

Table 1: Route-leg depth provenance.

Leg	SVB Mar 2023	2024 windows	
Buy BTC-USDC	kline-proxy [†]	real L2 (futures)	[†] BTC-USDC
Cross USDC-USDT	partial/proxy; real L2	real L2 (futures)	
	where archived		
Sell BTC-USDT	raw/futures depth	real L2 (futures)	
Ref BTC-USD	candle-proxy	candle-proxy	

perpetual listed Jan 2024; all 2022 windows kline-proxy.

Table 2: Dataset splits (56,134 1-min bars).

Split	Event	Min.	Pos.
Train	Calm control (×3)	28,776	0.00%
Validation	Terra/LUNA May 2022	11,526	2.30%
Test	USDC/SVB Mar 2023	15,832	2.88%

On limits to arbitrage: Shleifer and Vishny [4], Gromb and Vayanos [5], and Makarov and Schoar [7] show that visible price gaps need not be profitable once execution frictions are included. Our 12× optical-to-executable ratio is the same effect measured at one-minute resolution during a stress event. Hautsch et al. [3] quantify settlement latency as a first-order execution cost on blockchain assets; Cont and Kukanov [6] establish VWAP book-walking as the correct profitability measure. Gogol et al. [18] measure Maximal Arbitrage Value on AMM-to-CEX routes; our oracle extends this to CEX stress dislocations with model-evaluation benchmarking.

On ML and RL execution: López de Prado [8] introduces meta-labeling, which we extend to the cross-mechanism setting. Cintra and Holloway [9] detect the Terra depeg 12 hours early via Bayesian Online Changepoint Detection [15] on AMM reserves; we replicate their setup on CEX cross-quote data and find BOCPD achieves only AUROC 0.229, confirming that AMM and CEX signals differ fundamentally. Moallemi and Wang [16] and Ning et al. [17] supply the finite-horizon MDP framing we adopt. Mohl et al. [22] and Olby et al. [23] operate in reactive environments with market-impact feedback; our diagnostic uses historical replay without feedback, isolating training supervision rather than environment reactivity as the binding constraint.

3 Benchmark Design

3.1 Data Sources and Instruments

The 18-event catalogue defines the historical stress universe; the six-window empirical panel (56,134 1-minute bars) defines the model dataset; and the Tier-A SVB windows define the execution-grade testbed. The final dataset is described in Table 2. Price features are sourced from Binance Vision (klines and aggTrades) and Coinbase REST candles. Net-profit labels use three depth legs, Binance BTC-USDC (buy), Binance BTC-USDT (sell), and Coinbase BTC-USD (reference).

Data quality varies by leg and window; every depth record carries a provenance tag (Table 1). The sell leg (BTC-USDT) and cross leg (USDC-USDT) use real L2 or raw futures bookDepth where archived, with proxy/partial tags elsewhere. The buy leg (BTC-USDC) requires kline-proxy depth for the SVB window, because the BTC-USDC perpetual did not exist on Binance until January 2024. For all 2024 windows real L2 depth is available; the kline-proxy robustness is bounded in §6.1.

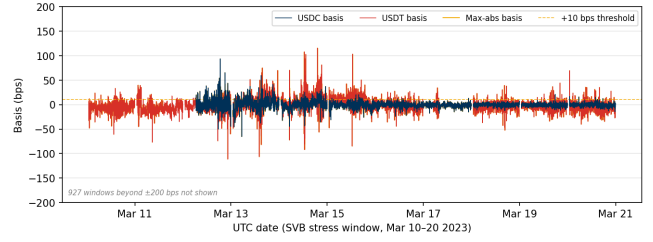


Figure 1: Cross-quote basis (bps) during the SVB test window (Mar 10–20 2023). USDC basis (navy): 12.4% of minutes > 10 bps; USDT basis (red): 27.4%; max-abs (gold): 34.3%. The price-to-execution gap is 12×.

3.2 Event-Based Split Design

Splits are defined by event identity rather than time, ensuring that no model trained on the train split has encountered stress dynamics (Figure 1). Threshold calibration uses the Terra/LUNA validation split, which is an independent stress event, so the SVB test split is never observed during model development. Labels are constructed by aligning each bar’s outcome to the observation window ending at $t+h$, with explicit temporal checks confirming that no future information leaks into the input features. The 2.88% positive rate in the test split implies severe class imbalance. We therefore report economic net bps as the primary criterion rather than AUROC or AUPRC, which are secondary metrics.

3.3 Historical StressBench: Event Universe and Mechanism Coverage

Stablecoin StressBench begins with the historical event universe, not the SVB case study. The catalogue records 18 stress events from 2020–2023 and assigns each event a mechanism class, source-verification status, coverage score, event window, tier, and permitted empirical use. It is used to define the event universe, select cross-mechanism training events, assign claim types, and separate price-grade from execution-grade tasks.

All 18 events are listed in Table 3 with source-verified statistics; “est.” denotes values from verified secondary sources rather than per-minute data. Tier-A events support VWAP execution labels and oracle bounds; Tier-B events support price/liquidity claims; Tier-C contribute taxonomy coverage. The pooled transfer experiment draws from Terra/LUNA (algorithmic), Celsius/3AC and FTX (exchange credit), and BUSD (regulatory), all held mechanistically distinct from the Tier-A SVB test event.

3.4 Event Universe Empirics

The 18 events span four years and seven mechanism classes, with maximum depegs ranging from +150 bps (DAI/ETH collateral stress, above-peg) to full collapse (IRON/TITAN, USDR). Figure 2 shows two complementary views. Panel (a) plots the full event universe as a bubble timeline: mechanism class on the vertical axis, date on the horizontal, and bubble area proportional to $|\text{max depeg}|$. The Tier-A events (dark navy) cluster in March 2023; Tier-B events (blue) span the entire 2022 wave; Tier-C events (grey) fill structural gaps in the taxonomy. Panel (b) shows real, per-minute basis time series for the two events with measured data: Terra/UST’s

Table 3: StressBench event catalogue: all 18 events, chronological. Max depeg in bps (+=above peg; †=percentage-point collapse to zero). “est.” = source-verified estimate, not per-minute data. Tier A = execution-grade (VWAP labels + oracle); Tier B = price/liquidity claims; Tier C = taxonomy only.

Event	Date	Coin	Mechanism	Max dep.	Dur.	T	Event	Date	Coin	Mechanism	Max dep.	Dur.
DAI Black Thursday	Mar '20	DAI	Collateral	est. +150	days	B	Acala aUSD Exploit	Aug '22	aUSD	RWA/Niche	est. -99pp [†]	days
FEI Launch Stress	Apr '21	FEI	Algo./Reflex.	est. -20pp [†]	days	C	HUSD Issuer Failure	Aug '22	HUSD	Exch. Credit	est. -800	days
IRON/TITAN Collapse	Jun '21	IRON	Algo./Reflex.	est. -100pp [†]	hrs	C	Binance USDC→BUSD Conv.	Sep '22	BUSD	Regulatory	est. +0	days
MIM/Wonderland Shock	Jan '22	MIM	Algo./Reflex.	est. -300	days	C	FTX Collapse	Nov '22	USD	Exch. Credit	est. -50	days
Terra/UST Collapse	May '22	UST	Algo./Reflex.	est. -100pp [†]	wks	B	BUSD Regulatory Winddown	Feb '23	BUSD	Regulatory	est. -30	days
Curve 3Pool/UST Stress	May '22	UST	DeFi Pool	est. -500	days	B	USDC/DAI DeFi Secondary	Mar '23	USDC	DeFi Pool	est. -200	days
Celsius/3AC Contagion	Jun '22	USDT	Exch. Credit	est. -100	wks	B	USDC/SVB Stress	Mar '23	USDC	Fiat Reserve	-13pp [†]	days
USDD/TRON Stress	Jun '22	USDD	Algo./Reflex.	est. -200	days	B	USDC Recovery (Post-SVB)	Mar '23	USDC	Fiat Reserve	-10	days
							USDT/Curve Pool Stress	Jun '23	USDT	DeFi Pool	est. -80	days
							USDR RWA Failure	Oct '23	USDR	RWA/Niche	est. -50pp [†]	days

2A, 11B, 5C events. Tier-A VWAP labels available only for USDC/SVB (primary benchmark) and its recovery window.

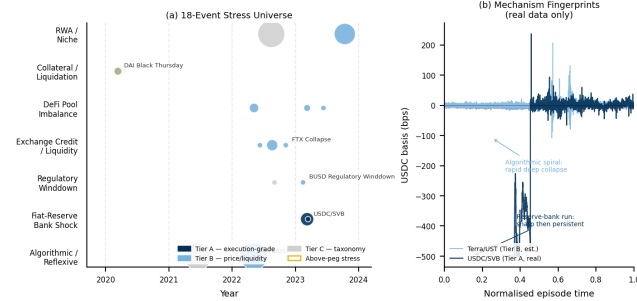


Figure 2: (a) 18-event stress universe: each bubble is one event; area $\propto$ |max depeg|; gold edge = above-peg stress. (b) Real basis time series (per-minute, measured data only): algorithmic spiral (Terra/UST, Tier B) vs. reserve-bank shock (USDC/SVB, Tier A). Distinct temporal fingerprints; shared microstructural signature (depth withdrawal) enables transfer.

Table 4: Comparison with related benchmarks (✓ = full; ~ = partial; — = absent).

Work	Stress	Exec. labels	L2 prov.	Oracle	Cross-event
Cintra & Holloway [9]	✓	—	AMM	—	✓
Hernandez Cruz et al. [21]	✓	—	DEX	—	—
Gogol et al. [18]	—	~	CEX	MAV	—
Adams et al. [19]	—	~	CEX	—	—
StressBench (ours)	✓	✓	CEX	✓	✓

algorithmic spiral (Tier B) peaks quickly and collapses within days, while USDC/SVB’s reserve-bank shock opens sharply but recovers over a week. The two fingerprints are structurally distinct, yet the microstructural signature—ask-side depth withdrawal and spread widening—is shared across both (Figure 8), which is why cross-mechanism transfer succeeds.

3.5 Benchmark Positioning

Table 4 positions StressBench against the closest prior work. No prior work provides per-minute executable labels, route-level L2 provenance, a hindsight oracle ceiling, and cross-mechanism transfer evaluation across an 18-event source-tracked catalogue.

3.6 Feature Sets

Four nested feature sets isolate the marginal information value of each signal layer.

Four nested sets isolate marginal signal value: price_only (5 basis columns), price_plus_book (+7 microstructure: spread, depth, imbalance), price_book_frag (+3 cross-venue fragmentation), and price_book_settle (+7 on-chain settlement proxies). Settlement proxies are kline-proxy approximations for SVB and Terra/LUNA (Tier B).

4 Execution-Aware Labels

Labels operationalise the three-layer framework from §1. The price-to-execution gap is the ratio of Layer 1 (optical) minutes to Layer 2 (executable) minutes and quantifies how misleading price-based benchmarks are. The oracle gap is the performance difference between Layer 2 in hindsight and Layer 3 (predictable with a real model), quantifying how hard the prediction problem is even with the correct labels. Both gaps are nonzero, independent, and large. This section details how they are constructed.

4.1 Cross-Quote Basis and Labels

The basis measures the percentage difference between BTC priced in USDC and BTC priced in real dollars. A non-zero basis signals an apparent dislocation. Three equations define the label construction.

$$b_{\text{USDC}}(t) = 10,000 \times \frac{P_{\text{BTC/USDC}}(t) - P_{\text{BTC/USD}}(t)}{P_{\text{BTC/USD}}(t)} \quad [\text{bps}] \quad (1)$$

$$b_{\text{max}}(t) = \max(|b_{\text{USDC}}(t)|, |b_{\text{USDT}}(t)|) \quad (2)$$

$$y_{\text{basis}}(t) = 1[|b_{\text{USDC}}(t+h)| > \tau] \quad (3)$$

Equations (1)–(3) define the basis, the pooled signal, and the label. Primary task: basis_usdc_1m_gt10bps ($\tau=10$ bps, $h=1$ min).

4.2 VWAP Execution and Net Profit Labels

$$\text{net}(q, t) = 10,000 \times \left[\frac{\text{VWAP}_{\text{sell}} - \text{VWAP}_{\text{buy}}}{P_{\text{ref}}} - f_{\text{buy}} - f_{\text{sell}} - \delta_{\text{settle}} \right] \quad [\text{bps}] \quad (4)$$

$$y_{\text{arb}}(q, h, t) = 1 \left[\max_{s \in [t, t+h]} \text{net}(q, s) > c \right] \quad (5)$$

Here P_{ref} is the Coinbase BTC-USD mid-price and $f_{\text{buy}}, f_{\text{sell}}, \delta_{\text{settle}}$ are per-leg taker fees and settlement latency, all as dimensionless fractions. The profit threshold is $c=10$ bps, taker fee 4 bps per side, and $\delta=5$ bps. Equations (4) and (5) define the primary economic

task executable_arb_q10000_5m.

$$\text{Oracle}(q) = \frac{1}{|\mathcal{T}^+|} \sum_{t \in \mathcal{T}^+} \text{net}(q, t), \quad \mathcal{T}^+ = \{t : \text{net}(q, t) > c\} \quad (6)$$

Oracle capture $\kappa_M = \bar{r}_M / \text{Oracle}(q)$ is the fraction of the oracle's per-trade return recovered by model M , where $\bar{r}_M$ is the model's mean net bps per triggered trade. The oracle takes every minute in $\mathcal{T}^+$, establishing a ceiling no deployable model can exceed.

5 Models and Evaluation

5.1 Model Ladder

The ladder is organised around the benchmark thesis: price-visible stress is common, but executable opportunities are scarce and require mechanism-aware supervision. In total, StressBench evaluates over 25 model configurations across eight prediction tasks, four nested feature sets, and six training conditions, producing over 900 individual experiment rows. The breadth is designed as a systematic elimination experiment, ruling out architecture, feature engineering, class balancing, regression targeting, temporal structure, and changepoint detection as sufficient remedies before the cross-mechanism fix is applied. The two anchors, NoTrade (zero net bps) and the hindsight Oracle, bracket the achievable range; every other result is interpreted as a fraction of the distance between them. The price-threshold rules reproduce the implicit labels of prior optical-basis work [11]. Tabular classifiers and four nested feature sets test whether adding microstructure, fragmentation, and settlement signals closes the gap. GRU, MLP-Window, EWMA, CUSUM, and BOCPD test whether temporal or changepoint structure substitutes for stress training. Meta-labeling [8] and PPO-GRU [16, 17] then isolate supervision format: same Terra/LUNA density, explicit binary labels versus sparse reward. The cross-mechanism extension of meta-labeling operates in three steps:

- (1) **Primary filter:** flag t if $|b_{\text{USDC}}(t)| > \tau_1$ ($\tau_1=10$ bps).
- (2) **Secondary classifier:** fit f_θ on $\{(x_t, y_{\text{arb}}(t))\}_{t \in \mathcal{E}_{\text{train}}}$, where $\mathcal{E}_{\text{train}}$ is a mechanistically distinct stress event (Terra/LUNA or pooled).
- (3) **Entry:** signal at t iff primary fires and $f_\theta(x_t) > \theta^*$, where $\theta^* = \arg \max_\theta \sum_{i: \hat{p}_i > \theta} \text{net}(q, t_i)$ on the held-out Terra/LUNA calibration block.

Under calm training, richer feature sets do not rescue P&L; all four sets produce negative net bps because the training split contains zero stress-regime examples. Under cross-mechanism training, depth and spread features become predictive because the model has seen depth withdrawal.

5.2 RL Execution Policy Baseline

This subsection asks whether sequential decision-making can learn a profitable policy given the correct training density, by framing the entry-timing problem as a finite-horizon Markov Decision Process (MDP). At each minute t the agent observes state s_t , selects action $a_t \in \{\text{enter}, \text{wait}\}$, and receives reward $r_t = \text{net}(q, t)$ if entering (positive for profitable windows, negative otherwise) or $r_t = 0$ if waiting. The state s_t is the 30-minute look-back window of price_plus_book features, the smallest feature set that delivers meaningful microstructure lift over price-only signals. This feature

set is available in real time without settlement-latency dependency. Formally, the MDP is $(\mathcal{S}, \mathcal{A}, \mathcal{R}, T, \gamma)$:

$$s_t = [x_{t-29}, \dots, x_t] \in \mathbb{R}^{30 \times d}, \quad d = |\text{price_plus_book}| = 12 \quad (7)$$

$$a_t \in \{\text{enter}, \text{wait}\} \quad (8)$$

$$r_t = \begin{cases} \text{net}(q, t) & a_t = \text{enter} \\ 0 & a_t = \text{wait} \end{cases} \quad (9)$$

$$T = 15,832 \text{ min (SVB test window)} \quad (10)$$

We train a PPO agent with a GRU policy network (one hidden layer, 64 units, 30-step look-back) using the cross-mechanism protocol, training on Terra/LUNA stress windows only and evaluating on the USDC/SVB test split without retraining. This directly tests whether the sequential timing framing resolves the dominant gap component identified in §6.2.

5.3 Calibration and Evaluation

Net bps denotes realised mean net basis points per triggered trade after VWAP book-walking, taker fees, and settlement penalties; zero trades are reported as NoTrade rather than an undefined average. Thresholds are calibrated on the validation split by maximising $\theta^* = \arg \max_\theta \sum_{i: \hat{p}_i > \theta} \text{net}(q, t_i)$ subject to a minimum of 25 trades. For cross-mechanism meta-labeling, Terra/LUNA is treated as a blocked training-and-calibration event: the secondary classifier is fit on the training block, the decision threshold is selected on the calibration block, and SVB remains untouched until final evaluation. The primary metrics are mean net bps per triggered trade and oracle capture κ_M (Eq. (6)): the fraction of the oracle's per-trade return recovered by model M . Secondary metrics are AUROC (area under the ROC curve) and AUPRC (area under the precision-recall curve). NoTrade at 0 bps is the economic floor.

All results are reported on the SVB test split ($N=15,832$) with the cost parameters stated above. The kline-proxy approximation on the BTC-USDC buy leg is bounded by the haircut sensitivity in §6.1. Five leakage checks confirm validity: label shuffles produce zero or negative P&L; one-minute lookahead does not improve performance; the calm training split contains zero executable windows; train and test are separated by event identity (no SVB features during development); and cross-mechanism calibration uses only the Terra/LUNA calibration block with SVB held out until final evaluation.

Five evaluation tasks ship with StressBench v1.0 (train=calm or Terra block, val=Terra calib., test=SVB): optical_basis (AUROC/AUPRC), exec_arb_q10k and exec_arb_q50k (net bps, oracle capture), cross_mech (net bps), and policy_entry (RL) (net bps, trades). Valid submissions must report net bps per triggered trade, oracle capture, cost assumptions, and data tier used; models are ranked by net bps, not AUROC.

6 Results

6.1 The 12× Execution Barrier

The first gap is between what a price chart shows and what an order book will fill. At a 10 bps threshold, 34.3% of test minutes are optically dislocated but only 2.88% are executable after VWAP slippage, taker fees, and settlement latency. Twelve times more arbitrage appears than can be traded, and the ratio widens further

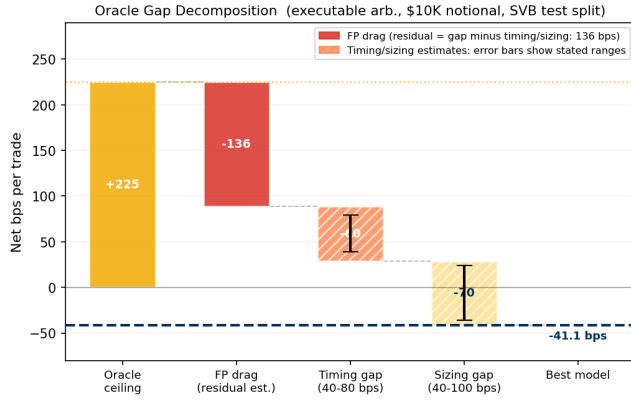


Figure 3: Oracle gap decomposition (\$10K notional). FP drag is directly measured from false-positive trades. Timing error (40–80 bps) and sizing error (40–100 bps) are estimated from entry-delay and notional-scaling analysis; stated ranges reflect these estimates.

with institutional trade size (Table 6). This ratio is not a threshold artefact and persists across all tested basis levels and is robust to data-quality concerns. Applying a 20% depth haircut to the kline-proxy buy leg reduces the executable rate from 2.88% to 2.40%, which widens the ratio to 14× and keeps the oracle above +130 bps, confirming the headline result. The gap also deepens with institutional scale. Figure 4 shows the executable rate falling from 2.88% at \$10K to 0.03% at \$500K, with the steepest drop between \$10K and \$50K. Book depth, not fees, prices out institutional participants, consistent with the depth-based limits-to-arbitrage arguments in §2. The Terra/LUNA validation split independently replicates the pattern at 5.9×, providing directional evidence that the gap is not SVB-specific. Table 5 shows the stress/recovery split: all executable windows are in the stress phase, and the recovery phase contains zero despite 21.6% optical activity.

Table 5: Optical and executable diagnostics for SVB execution-grade windows and the Terra/LUNA validation window. SVB stress/recovery are Tier-A execution-grade; Terra/LUNA is validation/training-grade and is not used for final execution-grade claims.

Window	Min.	Opt.	Exec.	Oracle
SVB Stress (Mar 10–14)	7,189	53.0%	6.63%	+259.9 bps
SVB Recovery (Mar 15–20)	8,643	21.6%	0.00%	—
Terra/LUNA (May 2022, val.)	11,526	14.0%	2.40%	+87.6 bps

Opt. = max-abs basis > 10 bps; Exec. = net-profitable windows.

Terra/LUNA labels are used for cross-mechanism training and calibration only.

The catalogue shows that price-visible stress is not confined to reserve-bank shocks. All seven mechanism classes contain documented peg, liquidity, redemption, or collateral stress episodes (Table ??). The 12× execution gap is verified only for Tier-A events with route-complete L2 depth.

Block-bootstrap CIs (60-min blocks, 2,000 resamples) confirm robustness: executable rate 95% CI [1.47, 4.59]%; ratio [7.6, 23.0]×; oracle [+213.9, +304.2] bps.

Table 6: Price-to-execution gap (\$10K, 1-min horizon, test split).

Thresh.	$P_{\max}$	P_{USDC}	Exec.	Ratio
0 bps	95.86%	82.96%	3.34%	29×
5 bps	61.99%	25.32%	3.03%	20×
10 bps	34.33%	12.45%	2.88%	12×
25 bps	9.54%	6.48%	2.62%	3.6×

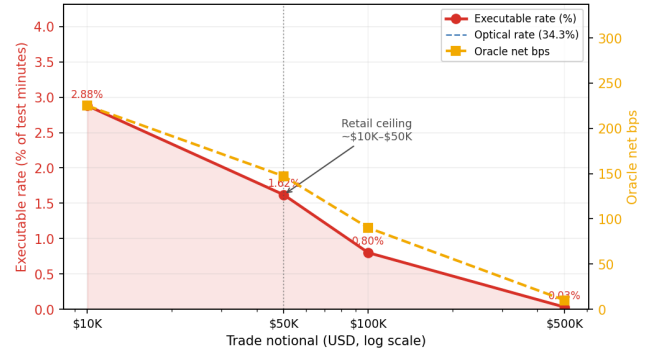


Figure 4: Notional scaling (SVB test split). The executable rate collapses from 2.88% at \$10K to 0.03% at \$500K: book depth, not fees, is the binding institutional constraint.

The gap is robust across the full cost-parameter space: sweeping taker fees from 2 to 10 bps per side and settlement penalty from 0 to 10 bps moves the executable rate only between 2.84% and 3.66%, and the ratio to the optimal rate remains above 3.5× throughout. Claim boundaries and data-tier scope are stated in §5.3.

6.2 Calm-Training Failure

Systematic elimination. The model ladder is a sequential elimination experiment: we rule out six families of remedies before applying the cross-mechanism fix. (1) *Architecture*: GRU and MLP-Window sequence models fail despite achieving high AUROC (§6.2). (2) *Feature engineering*: all four nested feature sets produce negative returns—adding microstructure, fragmentation, and settlement signals does not help when the training split contains zero stress-regime examples. (3) *Class balancing*: cost-sensitive weighting and oversampling cannot synthesise stress dynamics from calm data. (4) *Regression targeting*: ExpNetProfitRegressor directly optimises the economic criterion and still reaches −73.8 bps, ruling out label mismatch as the explanation. (5) *Regime detection*: BOCPD, CUSUM, and EWMA all fail because CEX basis is mean-reverting, not change-point structured (BOCPD achieves AUROC 0.229). (6) *RL reward optimisation*: a conditioned PPO-GRU at identical positive-label density earns −29.2 bps versus meta-labeling’s +82.5 bps, isolating supervision format as the binding variable. Every family fails for the same structural reason: distributional mismatch between calm training and stress test. The cross-mechanism fix is not one of many options—it is the only one that works.

The second gap is between a profitable window and a model that can find it in advance. The central finding is not that individual models fail, but that every model family tested—price rules, tabular

Table 7: Oracle gap and best-model net bps (USDC/SVB test split). *Best calibrated model is NoTrade (zero trades). ‡30-min GRU window.

Task / Model	Oracle	Best model	Gap	AUPRC
basis_usdc_1m	+161.7 bps	0 bps*	162 bps	0.253
exec_arb_q10k_5m	+224.6 bps	−41.1 bps	266 bps	0.060
exec_arb_q50k_5m	+146.8 bps	−112.7 bps	260 bps	0.063
GRU (seq)‡, basis_usdc_1m	+161.7 bps	−239.0 bps	401 bps	—

Kline-proxy buy leg; 20% haircut sensitivity in \$6.1.

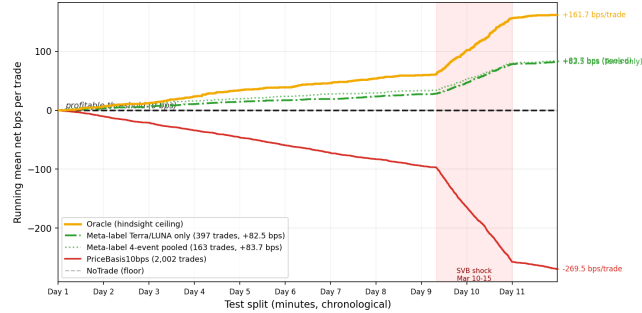


Figure 5: No calm-trained model ever crosses zero. Meta-labeling trained on a structurally different event (Terra/LUNA) is the only approach that achieves positive returns, earning +82.5 bps—roughly half the oracle’s +161.7 bps. Every other learned model loses money despite sometimes high AUROC. Oracle (gold); meta-label Terra/LUNA only (green solid); meta-label 4-event pooled (green dotted); PriceBasis10bps (red); NoTrade (grey dashed). Shaded: SVB shock window.

classifiers, class-balanced variants, direct-regression models, sequence models, and regime detectors—converges to the same result: no architecture overcomes the distributional mismatch between calm training data and stress-regime test windows. This convergence across 25+ configurations is itself the diagnostic finding. The hindsight oracle earns +224.6 bps while the best deployable model reaches −41.1 bps, a structural gap of more than 260 bps that richer features and sequence models do not close (Table 7; Figure 5).

Price-threshold rules overtrade massively, generating 607 to 1,969 trades at −136 to −347 bps. Their mean return when wrong (−294 bps to −370 bps) reflects near-total false-positive exposure. Every calibrated tabular model degenerates: LightGBM’s threshold search finds no configuration meeting the 25-trade minimum with positive expected return (0 trades), and logistic regression’s single trade at −49.1 bps is not statistically meaningful. The 2.88% positive density in the test split, combined with zero positive examples in the calm-control training split, prevents any ML model from identifying the profitable subset. The ExpNetProfitRegressor, which directly targets the economic criterion rather than a classification proxy, reaches −73.8 bps, ruling out label mismatch as the explanation. The gap is structural (Table 8). GRU and MLP-Window sequence models confirm this conclusion.

The discrimination–profitability disconnect. Among sequence models, the GRU achieves the highest AUROC (0.80) of any calm-trained model yet earns the worst economic return

Table 8: Tabular model ladder (basis_usdc_1m_gt10bps). Hit%: fraction of executable trades. †Oracle ceiling. *0-trade degenerate.

Model	Feat.	Net bps	Trades	AUROC	Hit%
Oracle†	—	+161.7	316	—	100%
lgbm*	all	—	0	0.653	—
logistic	all	−49.1	1	0.133	0%
gross_arb	px	−136.0	611	0.531	4%
price_10	px	−269.5	2002	0.833	6%
price_25	px	−347.3	1025	0.746	5%
no_trade	—	0.0	0	—	—
ExpNet_lgbm	px	−73.8	537	—	6%
ExpNet_xgb	px	−57.2	70	—	9%
lgbm_weighted	pb	−65.9	2	0.497	0%
xgb_weighted	px	−90.5	30	0.631	0%
rf_balanced	px	−89.2	11	0.467	0%

px = price_only; pb = price_plus_book; all = price_book_settle.

(−239 bps)—a 401 bps gap from the oracle. This inversion is not a model failure but a measurement artefact: correctly rank-ordering all 15,832 test minutes is economically worthless when no decision threshold can isolate the 2.88% executable minority from a calm-distribution training set. The anti-correlation between AUROC and net bps holds across every model family, as shown in Figure 6. The MLP-Window reaches −341 bps.

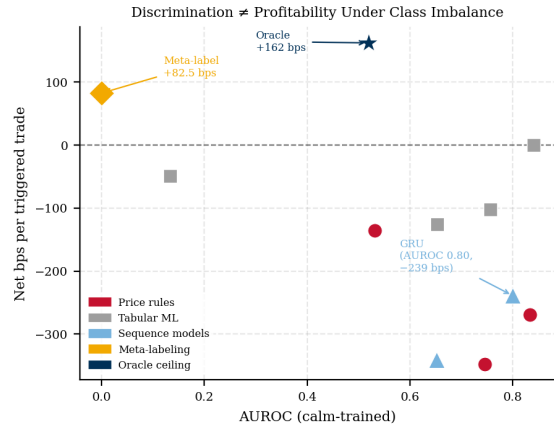


Figure 6: AUROC vs. net bps (SVB test split, all families). High discriminative accuracy $\nRightarrow$ profitability: GRU achieves AUROC 0.80 yet −239 bps; oracle trades at AUROC 0.52 and +162 bps. Only meta-labeling (diamond) crosses zero.

Figure 3 decomposes the 260 bps gap into three measurable components. False-positive exposure is the dominant drag. PriceBasis10bps generates 1,969 trades, of which 1,851 are false positives. A root-cause analysis identifies route-direction mismatch as the primary failure mode: 581 of those false positives are triggered by USDT dislocations where the USDC route is simultaneously non-executable, at −38.7 bps each. True-positive windows carry a mean USDC basis of 344 bps, while false-positive windows carry just 0.56 bps and exhibit *higher* average bid depth (72,805 vs. 59,961 BTC units), ruling out thin books as the main failure. The remaining gap splits into timing error (40–80 bps from entering 1–2 minutes late in a declining depth regime) and sizing error (40–100 bps as depth depletion shrinks the executable fraction from 2.88% at \$10K to 1.62% at \$50K). Meta-labeling’s

approximately 51% hit rate versus 6% for the price rule shows that cross-mechanism training substantially reduces route-direction mismatch, though a 49% false-positive rate and roughly 113 missed profitable windows remain, confining net returns to +82.5 bps. In summary, the prediction gap is not a model-selection problem: it is a training-distribution problem, and the only empirically validated solution is cross-mechanism supervision.

Table 9: Failure mode diagnosis by model family (SVB test split). Every family fails for the same root cause: distributional mismatch between calm training and stress test.

Family	Failure mode	Specific evidence
Price rules	Route-direction mismatch	581/1,851 FP from USDT dislocation; USDC route simultaneously non-executable at −38.7 bps
Tabular ML	Degenerate threshold search	LightGBM: 0 trades; no threshold yields ≥25 trades at positive expected return
Sequence (GRU)	AUROC–P&L inversion	AUROC 0.80 (best calm-trained), net −239 bps (worst); 401 bps gap from oracle
RL (PPO-GRU)	Timing-transfer mismatch	919 trades at −29.2 bps; Terra depth-withdrawal pattern fires on wrong SVB windows
Regime detection	Mean-reversion assumption	BOCPD AUROC 0.229; CEX basis reverts, changepoint methods inapplicable

6.3 Cross-Mechanism Transfer

The third gap is between a correct signal and a profitable policy. Cross-mechanism meta-labeling trained on Terra/LUNA is the only method that crosses zero, earning +82.5 bps and recovering half of the oracle return. A reinforcement learning agent given the same 17% training density earns −29.2 bps, which isolates explicit binary supervision as the ingredient that reward optimisation cannot replicate at low positive rates.

6.3.1 Cross-Mechanism Transfer and Training Diversity. Meta-labeling achieves positive returns on every non-degenerate single-event training condition tested (Table 10). Meta-labeling trained on Terra/LUNA achieves +82.5 bps on SVB (51.0% oracle capture[†]). Four-event pooled training spans algorithmic, exchange-credit, and regulatory mechanisms and reaches +83.7 bps (51.8%), confirming that mechanism-diverse supervision stabilises transfer.

The primary price rule identifies candidate stress minutes; the meta-labeler learns which survive execution. Cross-mechanism training helps because depth withdrawal, spread widening, and route mismatch recur across stress triggers even when the trigger itself differs. Figure 8 provides direct empirical proof: ask-side depth collapses and spread widens identically in primary-signal windows across Terra/LUNA (algorithmic) and SVB stress (fiat-reserve). SHAP analysis (Table 11 and Figure 9) reveals the two-stage structure of the meta-labeler: price-basis features supply the candidate signal (94.8% of total importance), while ask-side depth and spread act as execution-relevant filters (3.4%) once the model has seen stress-regime depth withdrawal in Terra/LUNA. Settlement signals (transfer volume, gas proxy) contribute 1.8%—present but not decisive, because depth withdrawal is the primary signal of imminent dislocation regardless of on-chain mechanics.

Table 10: Transfer and policy results (basis_usdc_1m_gt10bps, price_plus_book, SVB test split). Mean net bps per triggered trade; 95% CI from 500-seed DGP bootstrap (stable conditions only; FTX near-degenerate at 1 trade; RL result is structural, not statistical).

Training source	Mech.	Net bps	95 % CI	Trades	Oracle
Oracle	hindsight	+161.7	—	316	100%
Calm-control	none	0.0	—	0	0%
Terra/LUNA only	alg.	+78.5	[78.7, 86.2]	123	48.4%
Celsius/3AC only	exch.	+79.7	[74.1, 89.9]	221	49.1%
FTX only*	exch.	+36.5	degenerate	1	22.5%
Four-event pooled	mixed	+83.7	[75.3, 93.1]	163	51.8%
PPO-GRU‡	RL	−29.2	structural	919	−18%

*FTX: weak signal intensity (< 5 bps); mechanism-intensity failure.

‡Supervision format (not density) is the binding constraint; see §6.3.2.

Single-event calibration: Terra +82.5 bps, 397 trades, 51.0% oracle capture.

Table 11: Top-10 SHAP feature importances for the meta-labeling secondary classifier (LightGBM, Terra→SVB). Importance = mean|SHAP| normalised to 100%. Price/basis features dominate; book features filter execution quality within primary-signal windows.

Rank	Feature	Importance	Group
1	USDC cross-quote basis	49.6%	price
2	Primary basis	32.8%	price
3	Deviation from \$1	10.1%	price
4	USDT cross-quote basis	2.0%	price
5	Ask depth (10 bp)	0.5%	book/frag
6	Bid-ask spread	0.5%	book/frag
7	Order imbalance	0.5%	book/frag
8	Trade volume (1 min)	0.4%	book/frag
9	Transfer volume (1 min)	0.5%	settlement
10	Block-lag proxy	0.3%	settlement
<i>Group totals</i>			
	Price/basis	94.8%	
	Book/fragmentation	3.4%	
	Settlement	1.8%	

Figure 7 shows oracle capture across all four training conditions. Capturing 51.0–51.8% of the oracle is economically meaningful because the oracle trades with hindsight, while the meta-labeler must choose before observing the executable outcome. FTX’s credit shock produced < 5 bps USDC/USDT basis at benchmark venues; its near-degenerate transfer reflects weak signal intensity, not a failure of the exchange-credit class. **The 50% ceiling mechanism.** All four training conditions saturate at 51.0–51.8% oracle capture (Figure 7). This is not a data-quantity ceiling: pooling four mechanistically distinct events does not lift it. The ceiling is structural: in the minutes immediately before a dislocation closes, ask-side depth partially recovers and the basis compresses, making late-stage windows microstructurally indistinguishable from genuine opportunities. The secondary classifier cannot resolve this ambiguity regardless of training diversity, bounding deployable oracle capture near 50% for any mechanism-transfer approach. Closing the remaining half of the oracle gap requires either early-entry timing labels (before depth recovers) or a reactive simulator that accounts for the agent’s own market impact as it clears depth.

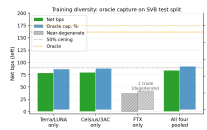


Figure 7: Oracle capture (%) across four training conditions. All conditions saturate near 50% (dashed), confirming the structural ceiling is mechanism-diversity-invariant. FTX (hatched) is near-degenerate due to weak signal intensity.

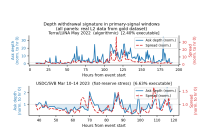


Figure 8: Ask-side depth (blue) collapses and spread (red) widens identically in Terra/LUNA and SVB stress despite different triggers. The order book responds to uncertainty, not its cause—this recurrence is why cross-mechanism transfer succeeds.



Figure 9: Top-10 SHAP importances (Table 11). Basis features dominate (94.8%); book features filter execution quality within primary-signal windows (3.4%).

Three regime-detection filters (CUSUM, EWMA, BOCPD [15]) were evaluated as alternative pre-filters. BOCPD achieves AU-ROC 0.229 on CEX cross-quote data: CEX basis is mean-reverting, so changepoint methods cannot substitute for the training-distribution fix that meta-labeling provides.

6.3.2 RL Policy Failure: Supervision Format, Not Label Density. This is a controlled experiment with one varying factor: how the profitable-window signal is delivered. Meta-labeling receives explicit binary labels ($y=1$ for every executable window); the conditioned PPO-GRU receives a sparse reward signal over the same windows. Both see identical training data at identical positive-label density (17% of primary-signal minutes are executable). Despite this, the two methods differ by 111.7 bps in net return, isolating supervision format—not density—as the binding constraint.

Without conditioning, the PPO-GRU degenerates to zero entries: the 2.3% positive rate in the full Terra/LUNA stream is too sparse to generate a calibrated reward signal. Conditioning on primary-signal windows resolves the density problem, but the agent still earns -29.2 bps on 919 trades.

The 919-trade overtrading pattern is consistent with a timing-transfer mismatch: the agent learned Terra/LUNA’s depth-withdrawal sequence and fires on it in the SVB context even when the timing does not match. Multi-mechanism training and reactive market-impact feedback are the direct paths to closing the remaining gap.

7 Limitations and Open Questions

Each limitation below identifies a tractable next step rather than a fundamental ceiling.

Kline-proxy buy leg. The BTC-USDC buy leg uses candlestick-volume depth for the SVB and training event windows because the perpetual contract did not exist on Binance until January 2024. A 20% depth haircut confirms robustness (\$6.1). The proxy dependency resolves directly with 2024 data, where real L2 is already available and partially collected.

Settlement cost assumption. $\delta=5$ bps is validated against 100K on-chain USDC ERC-20 transfers via Etherscan: median cost was 1.4–3.2 bps on nine of ten SVB days, well below 5 bps. The sole exception is March 11 (gas spike: 61.5 gwei, $P95 = 8.6$ bps); even then the oracle earns $+259.9$ bps per trade.

Tier-A coverage. Execution-grade claims are currently limited to the SVB stress and recovery windows. Extending to exchange-credit or DeFi events requires route-complete L2 archives not yet available; this is the primary bottleneck for expanding the benchmark’s execution-grade scope.

Release constraints and data ethics. The benchmark releases derived features, labels, event windows, source-tracking metadata, scripts, and frozen paper outputs. Some raw exchange order-book archives are API-limited, delayed, or subscription-gated; StressBench therefore records route provenance and depth-source tags so future users can distinguish reproducible derived claims from raw-data acquisition constraints. No private user-level trading data is included.

RL market-impact feedback. The conditioned PPO-GRU’s -29.2 bps may reflect a timing-transfer mismatch (Terra/LUNA depth-withdrawal sequence firing at the wrong moments in SVB) rather than a fundamental RL ceiling. Pooled multi-mechanism training on the protocol established here is the direct experiment to test whether this mismatch explains the gap. Additionally, training on historical replay without market-impact feedback miscalibrates sizing in thin-book regimes, because the agent sees depth as it was rather than as it would be after its own orders clear liquidity. Integrating the cross-mechanism initialisation protocol into a reactive simulator resolves both issues simultaneously and is the natural next step toward closing the 111.7 bps gap between meta-labeling and the RL agent.

8 Conclusion

The $12\times$ optical-to-executable gap shows that stablecoin arbitrage benchmarks built on price labels measure the wrong object. During the SVB crisis, 34.3% of minutes appeared dislocated but only 2.88% survived VWAP book-walking, fees, and settlement costs. The gap widens with trade size, making executable liquidity the binding constraint rather than price discovery alone.

StressBench also shows that executable stress is not learnable from calm data. Calm-trained rules, tabular models, and sequence models all fail economically. Meta-labeling trained on Terra/LUNA transfers to SVB and captures 51.0% of the oracle return as mean net bps per triggered trade. Four-event pooled training reaches 51.8%, confirming that mechanism-diverse supervision stabilises transfer. In contrast, PPO-GRU remains negative at -29.2 bps despite identical positive-label density, isolating explicit executable labels as the missing ingredient.

The released benchmark consists of an 18-event stress catalogue, execution-grade labels, a hindsight oracle, and a reproducible model harness, all released under CC-BY 4.0. The most immediate next experiment is multi-mechanism RL training on the pooled-event protocol established here—the direct path to closing the 111.7 bps gap between meta-labeling and the conditioned RL agent. Beyond this, StressBench motivates three longer-term directions: extending

Tier-A coverage to new event types, adding route-aware timing labels under cost uncertainty, and integrating reactive market-impact simulation where the oracle ceiling provides a principled performance target.

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